End-to-end user journey

Same pattern as the LLM-app architecture diagram: **user input on one side**, system processing in the middle, **user output on the other side**.

Pause | Prev | Next | Normal | Step 1 of 9

Step 1 of 9: User input analyst asks
End user (left) submits a natural-language risk/policy question with JWT session context.

? USER INPUT | SYSTEM PROCESSING ? | USER OUTPUT ?

USER INPUT

END USER
Investment / Risk Analyst

Query
Are any assets exceeding the allowed risk thresholds for our portfolio strategy?

Auth context
JWT role allowed_portfolio_ids

End user sends query to the system via UI.

INPUT ENRICHMENT & RESPONSIBLE AI

1. UI / FastAPI
POST /api/v1/query trace_id

?

2. Input guardrail
Block override / bypass / injection

?

3. Router
Classify C1C5 extract entities

? fan-out

4a. RAG enrich

Embed ? pgvector policy snippets

4b. SQL enrich
Templates ? finance_risk_db (+ RBAC filter)

?

5. Risk agent
Thresholds vs scores E1E8 check

?

6. Synthesis / Escalate
Evidence-only answer or human handoff

?

7. Output validator
Citations no advice disclaimer

USER OUTPUT

END USER
Same analyst receives result

Response envelope
Answer + DOC:/SQL: citations + risk summary + disclaimer

Or escalation
Refusal + handoff envelope (trigger, evidence, reviewer)

Output sent to end user.

1 **User input analyst asks**
End user (left) submits a natural-language risk/policy question with JWT session context.

2 **Enter the system UI / API**
Query moves from user input into FastAPI: authenticate, attach portfolio scope, stamp trace_id.

3 **Guardrail (responsible AI)**
Input guardrail screens override/bypass intent before any enrichment or LLM call.

4

Router chooses path

Query classified into C1C5; entities extracted so enrichment is scoped.

5

Input enrichment RAG + SQL

Parallel enrichment: policy snippets from pgvector and structured rows from SQL templates, RBAC-filtered.

6

Risk reconcile + escalation gates

Risk agent compares policy vs data. Any E2E8 trigger will divert to human handoff.

7

Synthesis or escalate

Clear path ? evidence-only synthesis. Triggered path ? escalation envelope (no autonomous answer).

8

Output validator

Citations required, no investment advice, disclaimer present then package for the user.

9

User output answer delivered

Result returns on the output side to the same end user: grounded answer + citations, or escalation status.